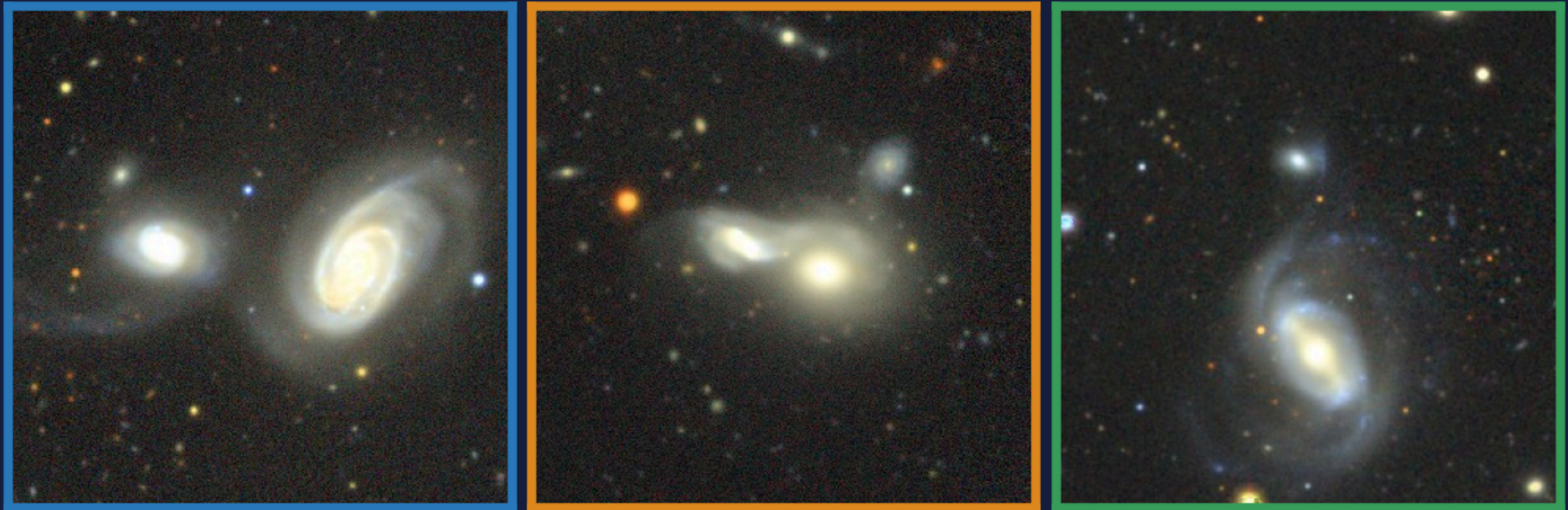


Active nuclei and the interbars sample

Is the gas in the nucleus of a barred, interacting galaxy lit by young stars — or by a black hole?



A practical analysis on already reduced spectra

LECTURE 1 · SLIDES

Active nuclei

Black holes, how an AGN is built, and why its light leaves a different mark on gas than starlight does.

LECTURE 1 · SLIDES

The sample

27 barred galaxies caught at three stages of an interaction, and the fibre spectra of their nuclei.

LECTURE 1 · NOTEBOOK

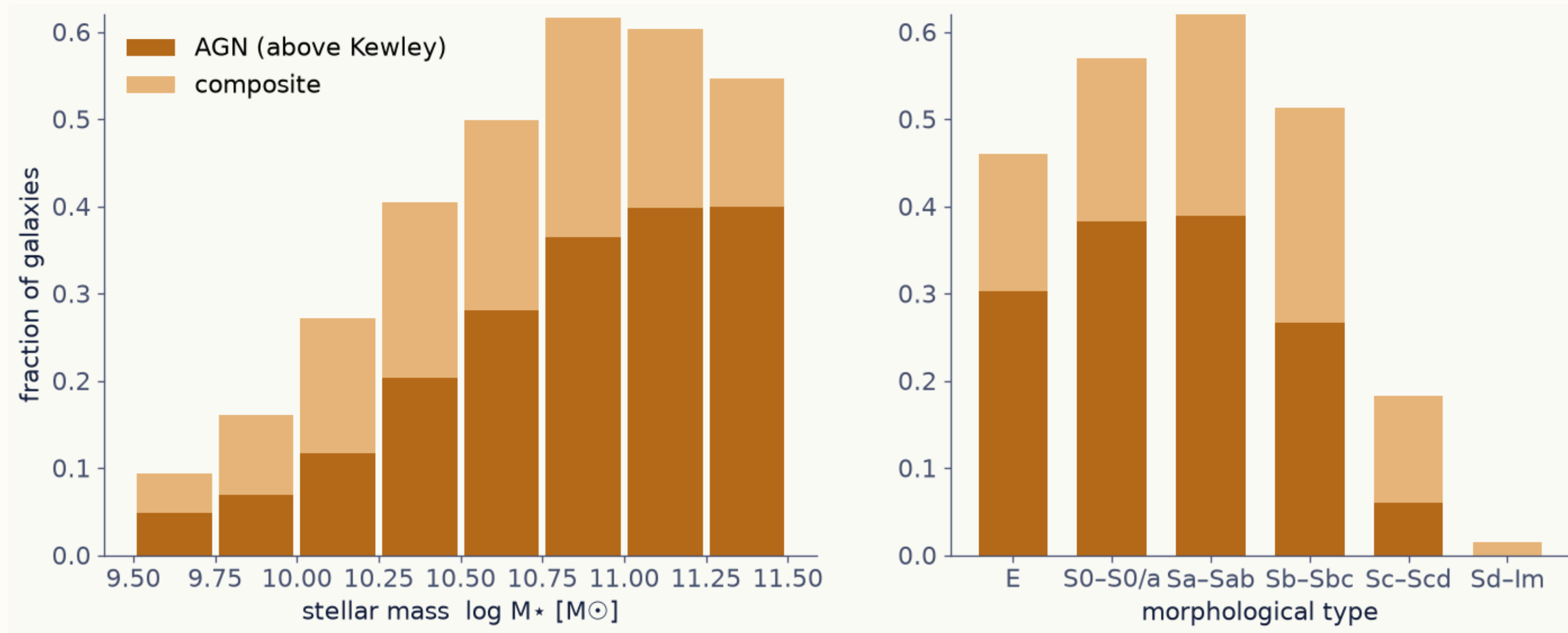
Spectra and pPXF

Read a real spectrum and learn how pPXF models it: templates for stars and gas, motions, dust.

LECTURE 2 · NOTEBOOK

Build mock galaxies by hand, then load the spectra of your galaxies, fit them with pPXF, and place them on the BPT diagram of barred and of interacting galaxies.

How common are AGN?

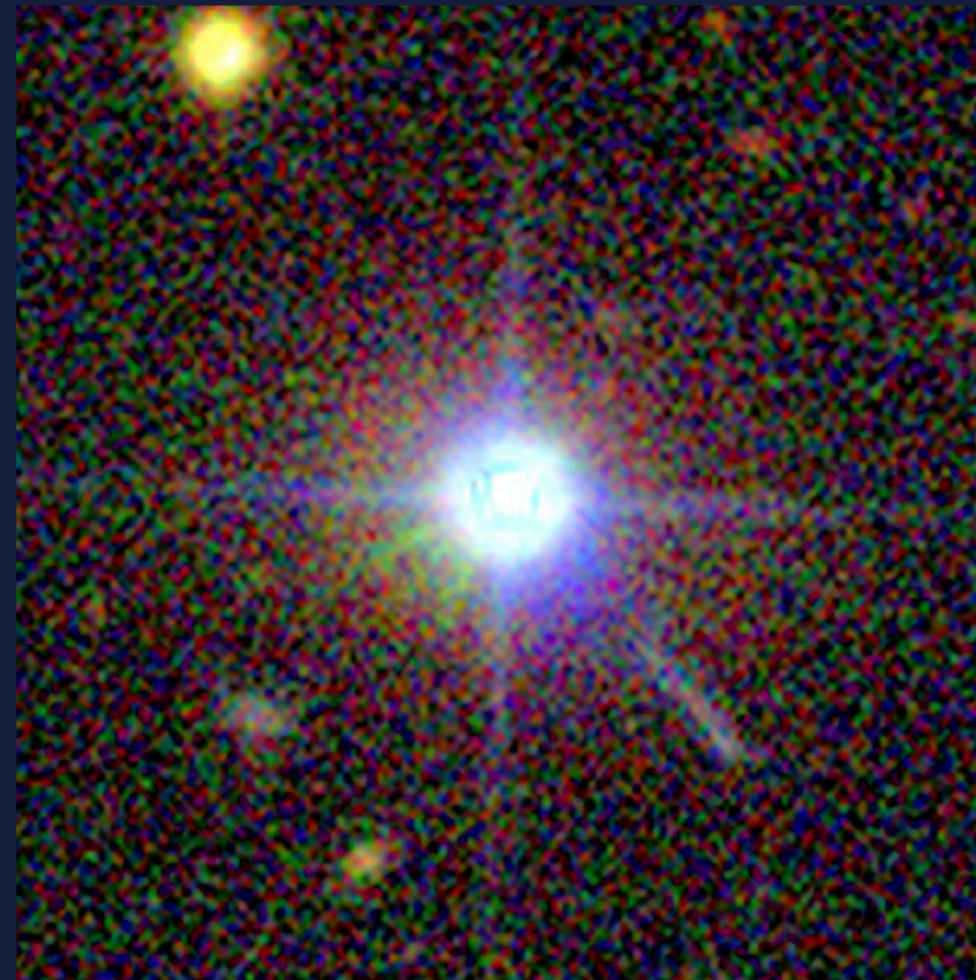


Active-looking nuclei **rise steeply with stellar mass** and live mostly in **bulge-dominated** galaxies (E to Sbc); they are rare in small, late-type discs.

A region smaller than the Solar System can outshine its whole galaxy

In very bright quasars the AGN can outshine its host galaxy by a factor of 10–100. On the other hand, in Seyfert galaxies the AGN might be hidden within the light of the galaxy bulge.

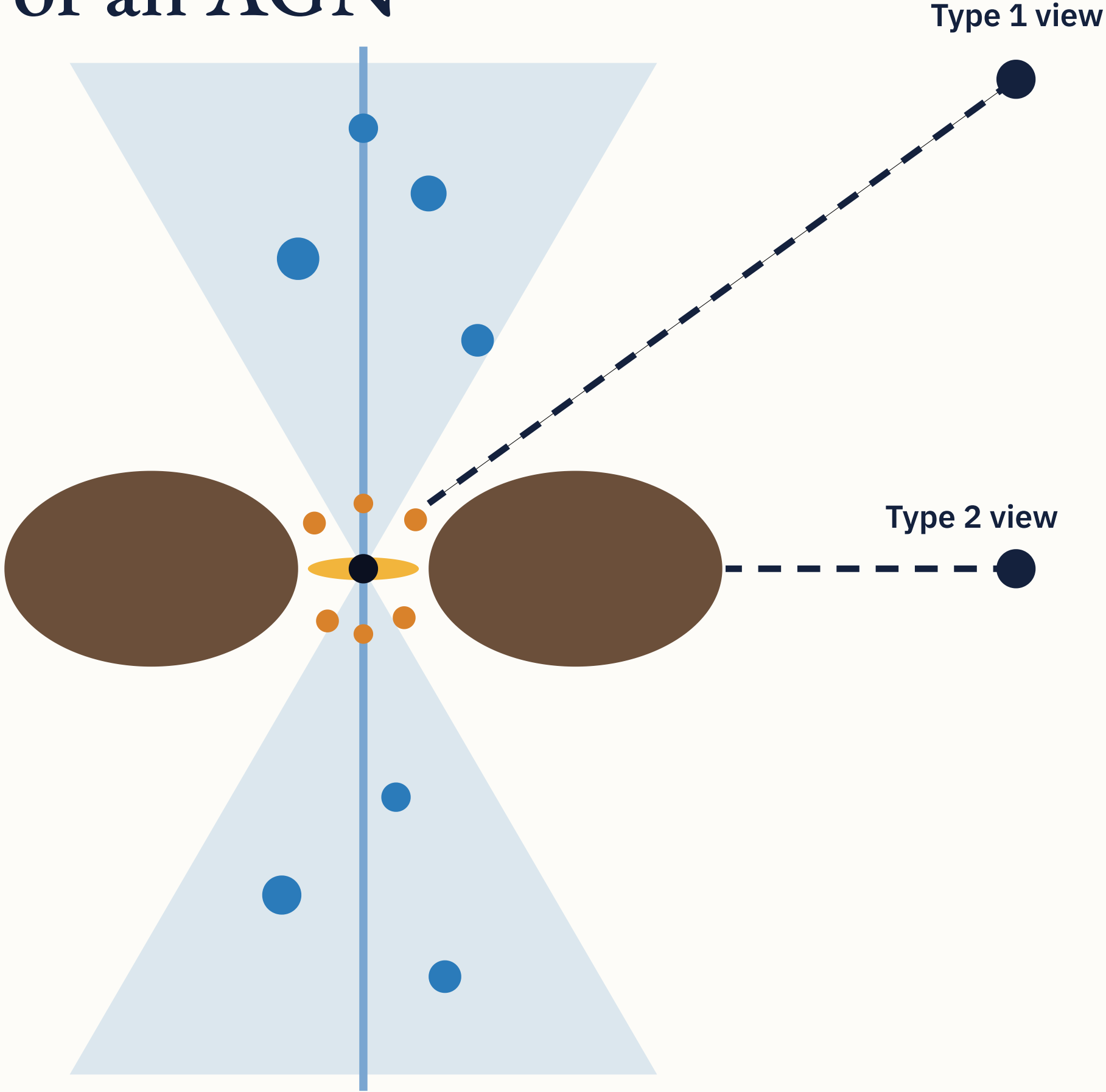
3C 273 — a quasar, $z = 0.16$



NGC 4151 — a Seyfert, 16 Mpc away



Anatomy of an AGN



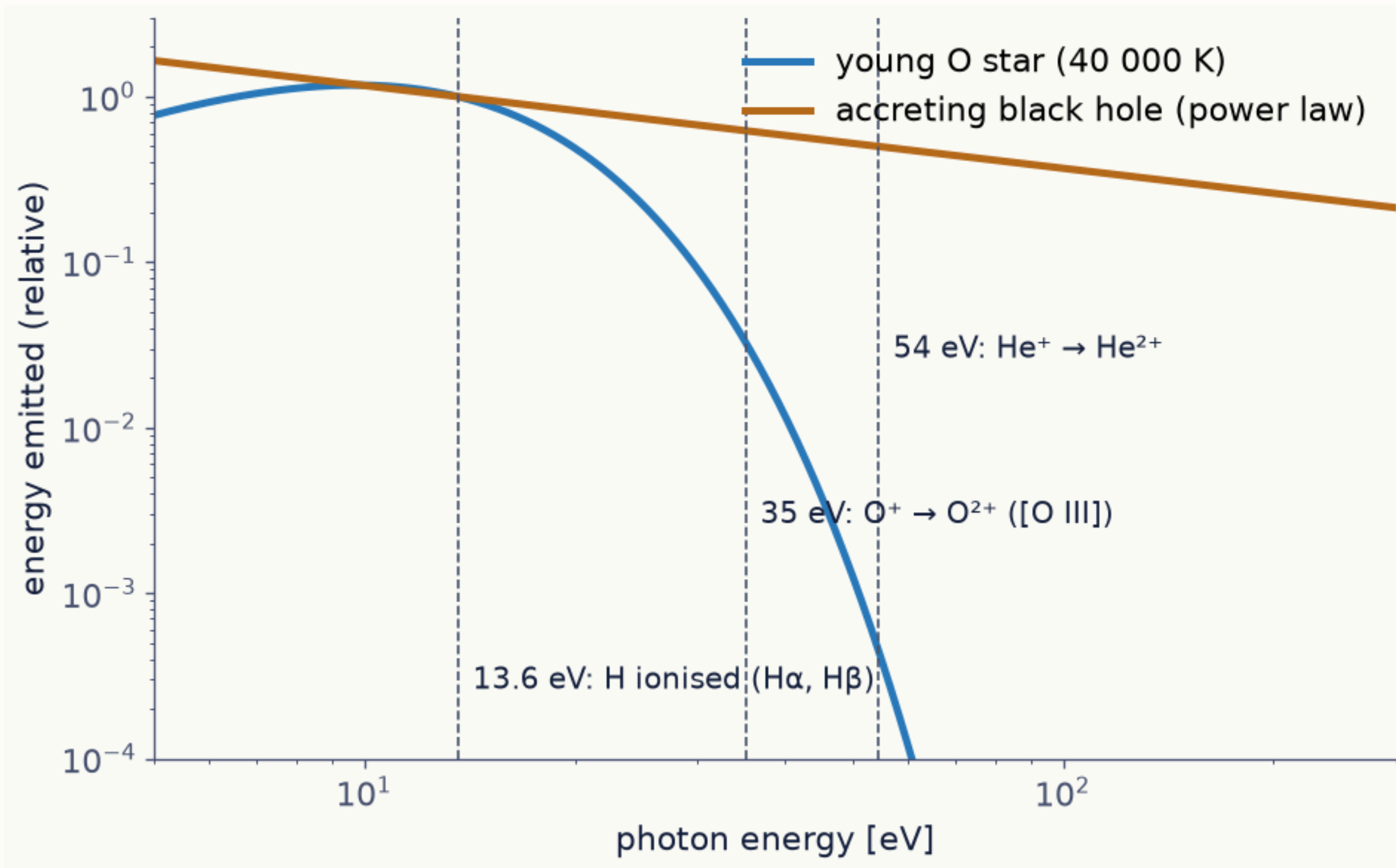
- **Black hole + accretion disc** — hot gas spiralling in; UV and X-ray light.
- **Broad-line region** — clouds within light-days, orbiting at thousands of km/s.
- **Dusty torus** — hides the centre when seen edge-on.
- **Narrow-line region** — gas 0.1–1 kpc out, lit through the cones; a few hundred km/s.
- **Jet** — present in some AGN (radio-loud).

Type 1 and type 2 - AGNs

	Type 1 (e.g. Seyfert 1)	Type 2 (e.g. Seyfert 2)
Line of sight	Above the torus, into the cone	Through the torus
Broad lines	Yes: H α , H β thousands of km/s wide	No: hidden by dust
Narrow lines	Yes	Yes: [O III], [N II] strong
How we spot it	The broad wings under H α	The ratios of the narrow lines: the BPT

The torus and the stellar disc do not need to lie in the same plane: even in a face-on galaxy we can be looking through the torus, and see a type 2 AGN.

Stars and black holes ionise gas differently



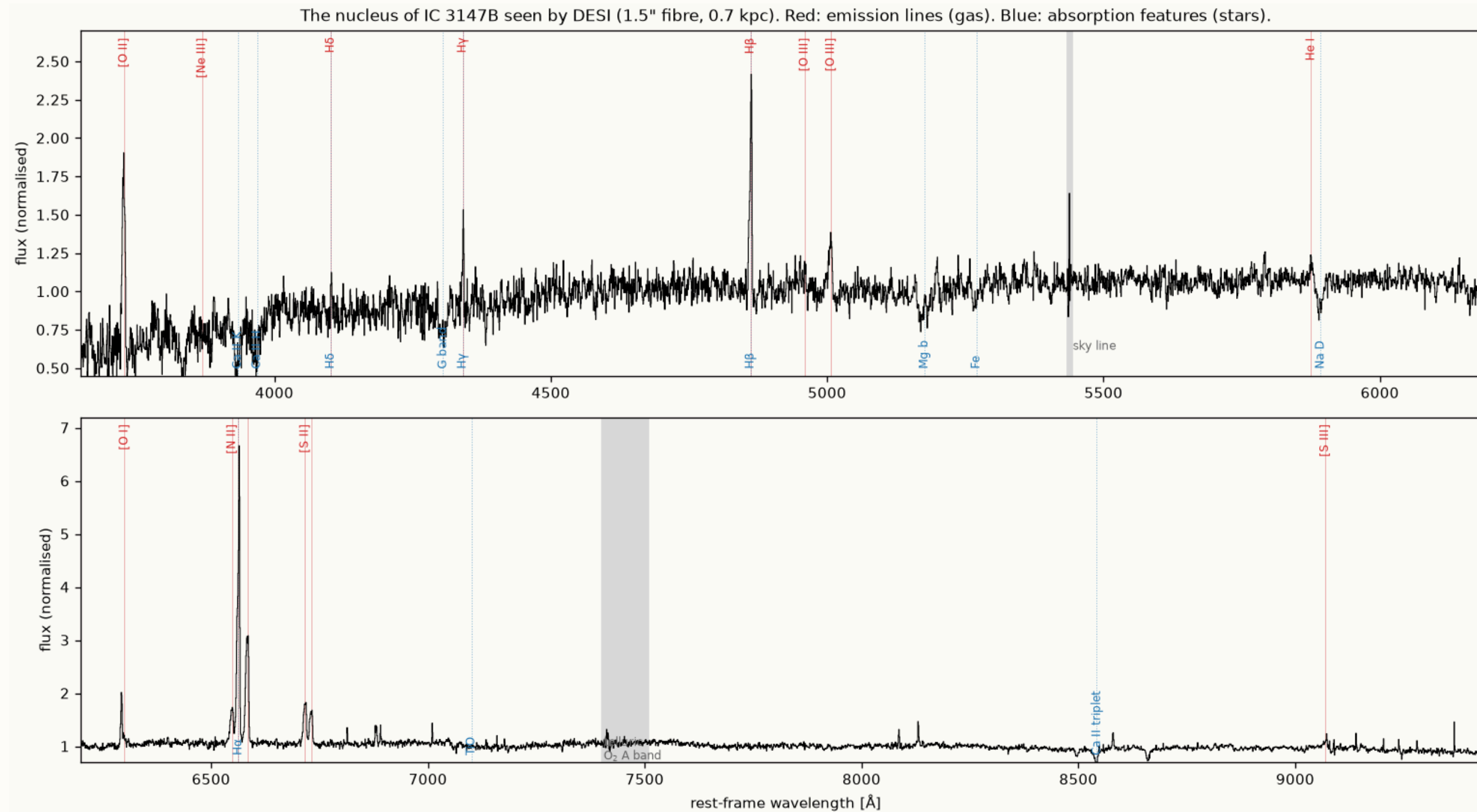
Young O and B stars

Few photons above 35 eV, and a sharp edge to the ionised gas. **H α , H β strong; [N II], [O III] moderate.**

An accreting black hole

Plenty of hard photons make O^{2+} ([O III]), and X-rays heat a large partly ionised zone. **[O III], [N II], [S II], [O I] strong.**

A spectrum is written by stars and by gas

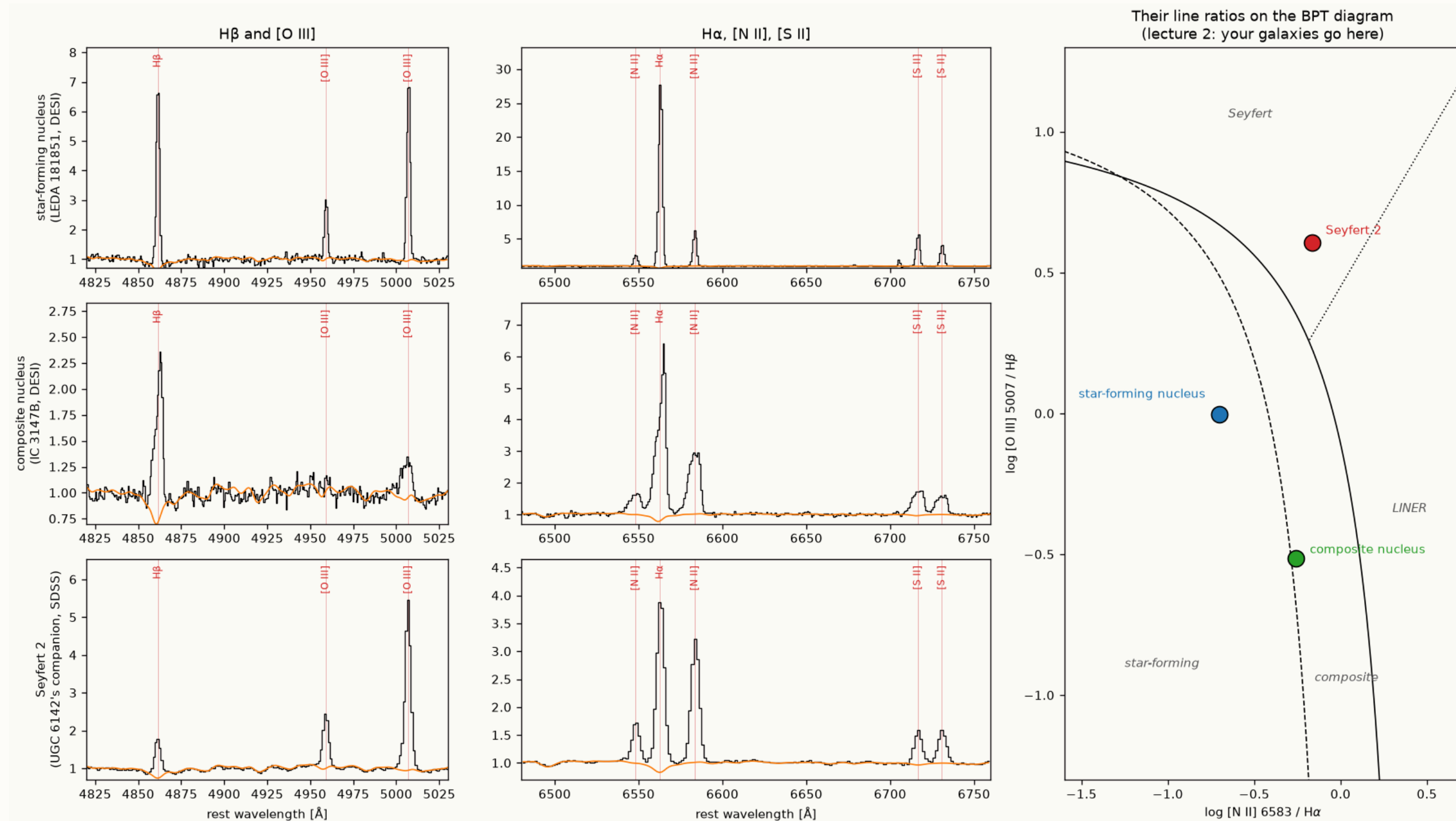


Red: emission lines from ionised gas.

Blue: absorption by stellar atmospheres.

The continuum: the summed light of $\sim 10^9$ stars.

The BPT diagram orders nuclei by what ionizes their gas



x: $[\text{N II}] / \text{H}\alpha$
y: $[\text{O III}] / \text{H}\beta$

Each ratio pairs lines close in wavelength: dust barely moves the point.

Dashed (Kauffmann 2003):
below, stars alone.

Solid (Kewley 2001): the
most stars can do. Above,
an AGN.

Getting gas to the nucleus

To reach the black hole, gas orbiting kiloparsecs out must lose almost all of its angular momentum. Something has to exert a torque on it.

Interactions

Tidal forces from a companion funnel gas to the centre. Close pairs in SDSS show more AGN than isolated galaxies of the same mass (e.g. Ellison et al. 2011).

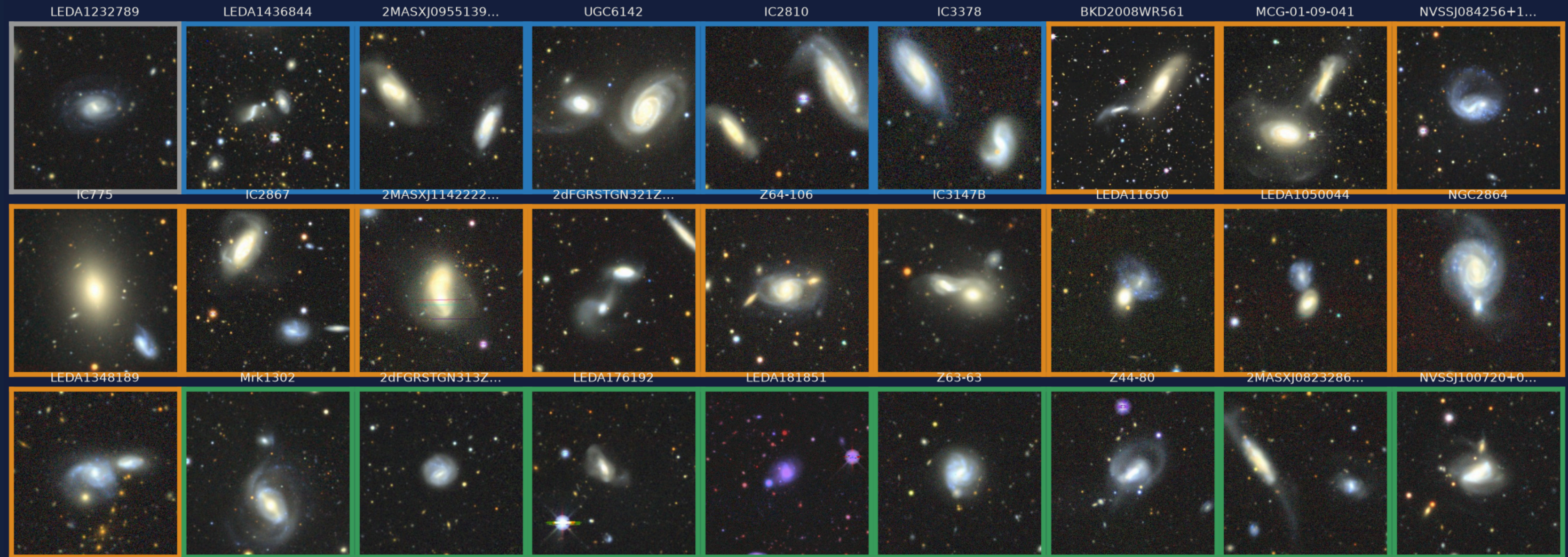
Bars

A rotating bar drives gas inwards along its leading edges, feeding central star formation. Whether it also feeds the AGN is debated (e.g. Ho et al. 1997; Cisternas et al. 2013).

What happens when a galaxy has both?

27 systems

Barred galaxies at $z = 0.007\text{--}0.045$, 18 with a named companion. Each image is 70 kpc across.



Interaction stage, visual classification: ■ 5 · pre-interaction ■ 13 · merger ■ 8 · post-merger ■ 1 no class

Three stages of an interaction



Pre-interaction

UGC 6142 - two separate galaxies

Merger

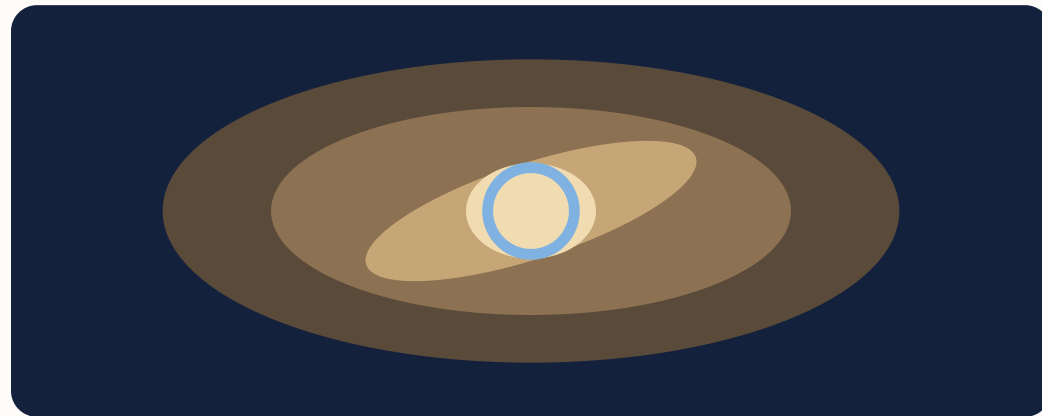
IC 3147B + IC 3147

Post-merger

Z44-80 - one remnant, tidal loops

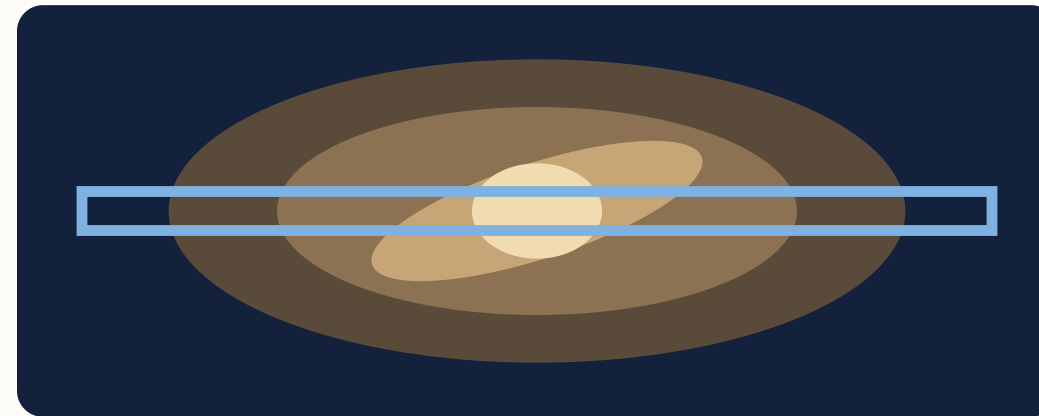
Three ways to take a spectrum of a galaxy

Fibre / aperture



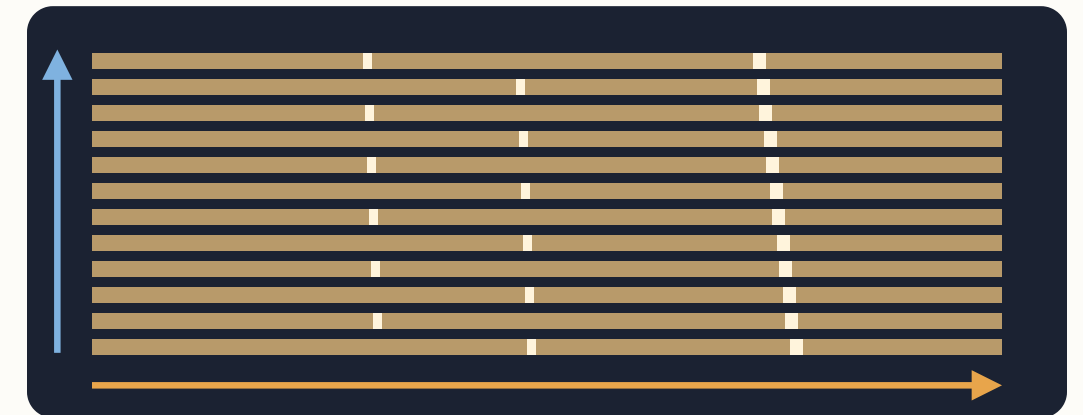
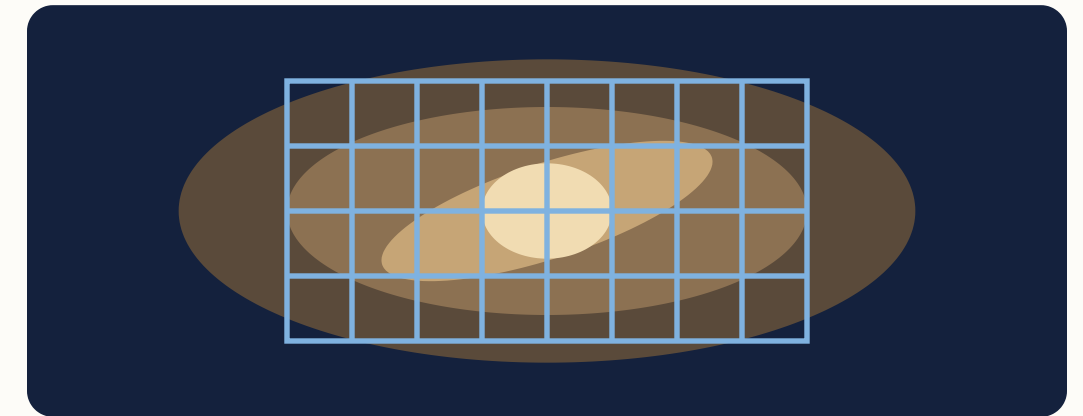
One spectrum of all the light inside the aperture. SDSS 3", DESI 1.5": **our data**.

Long slit

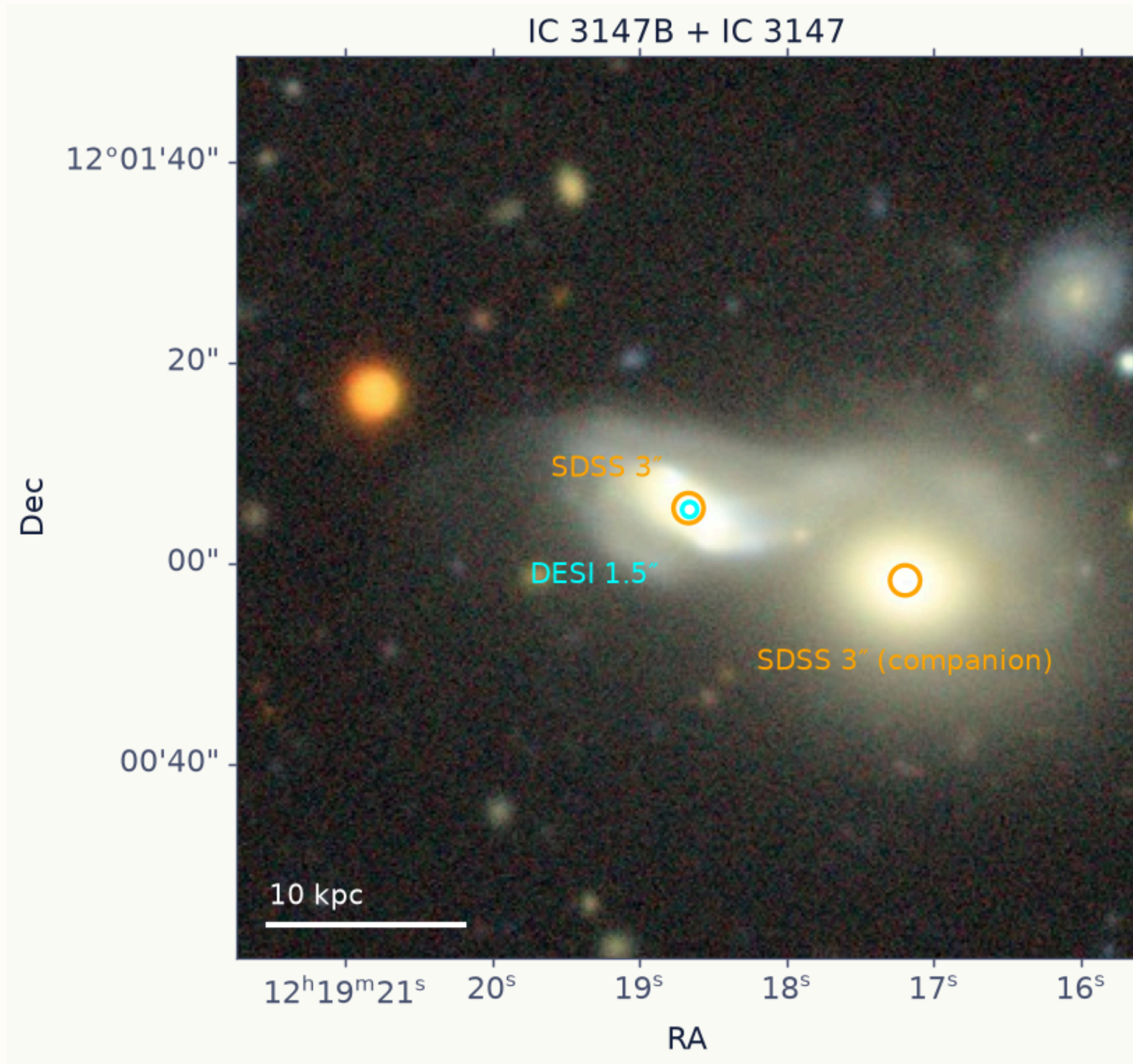


A spectrum at every position along the slit. Tilted lines = rotation: velocity versus radius.

Integral field (IFU)



A spectrum at every position of a 2D field: a data cube (x, y, λ). E.g. MaNGA, CALIFA, MUSE.



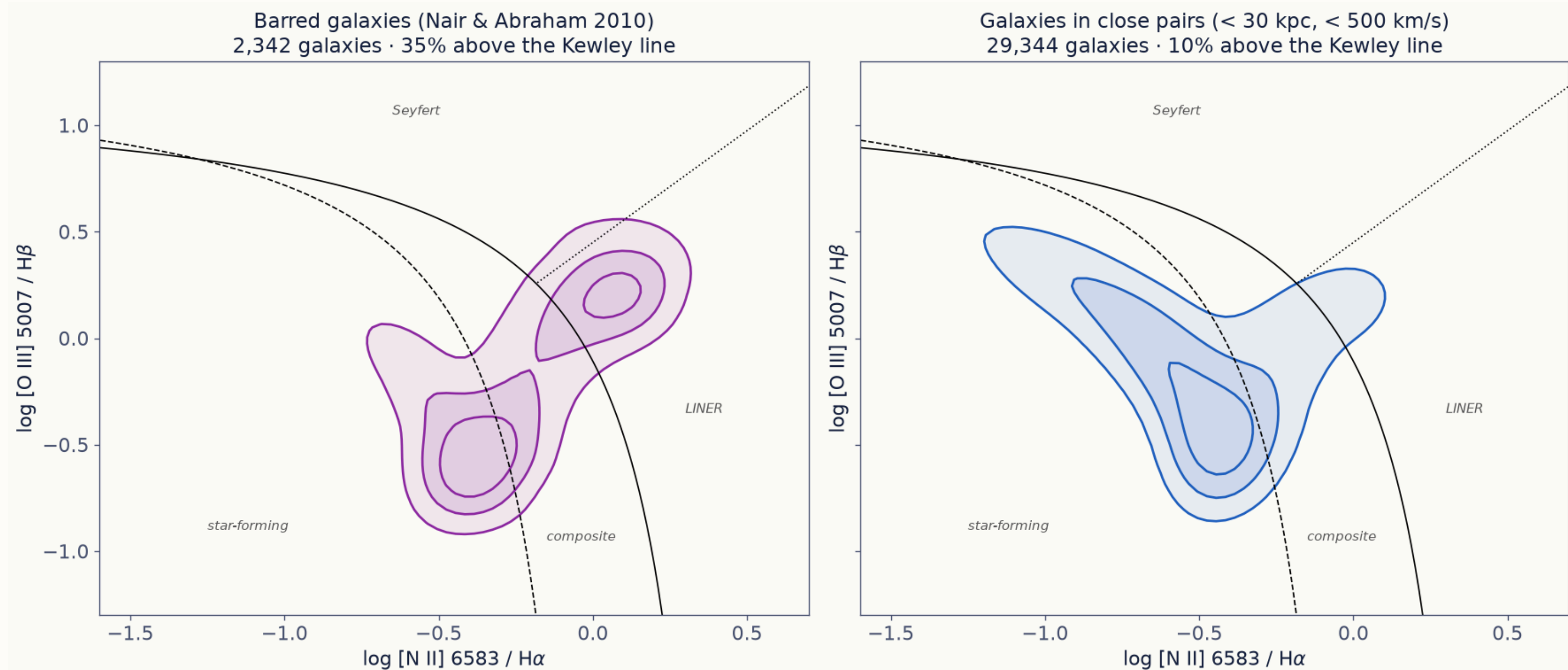
The data: fibre spectra of the nuclei

Survey	Fibre	Spectra
SDSS DR17	3"	40
DESI DR1	1.5"	12

52 spectra: 34 of the barred galaxies, 18 of their companions. Public, downloadable by anyone.

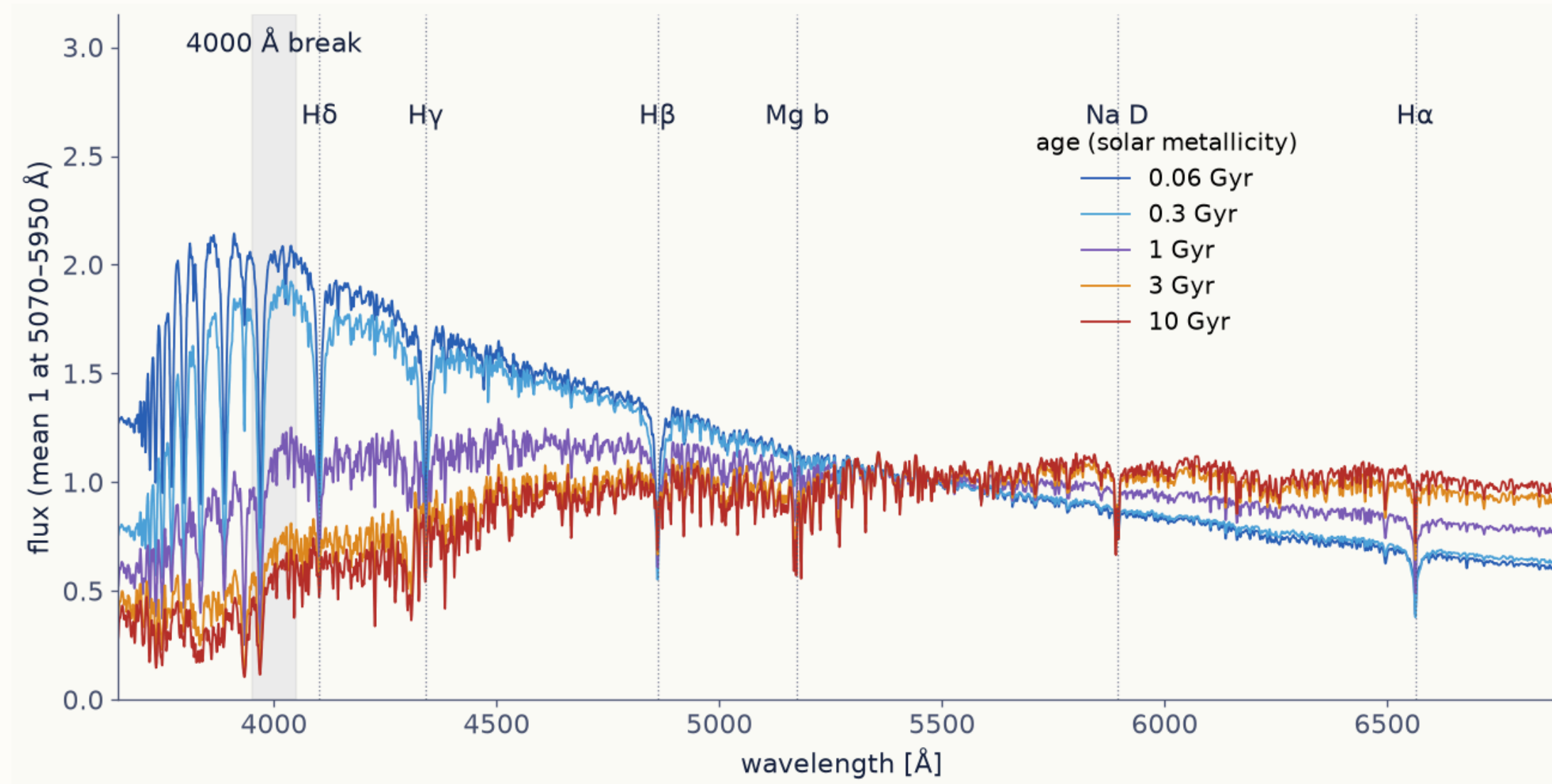
A fibre sees only the central 1–2 kpc: **the nucleus, not the bar.**

Our yardsticks: barred and interacting galaxies in SDSS



Contours enclose 20, 50 and 80 % of each population; line fluxes from the MPA-JHU catalogue. **In lecture 2 your galaxies go on these two diagrams.**

Stellar templates: simple stellar populations (SSP)



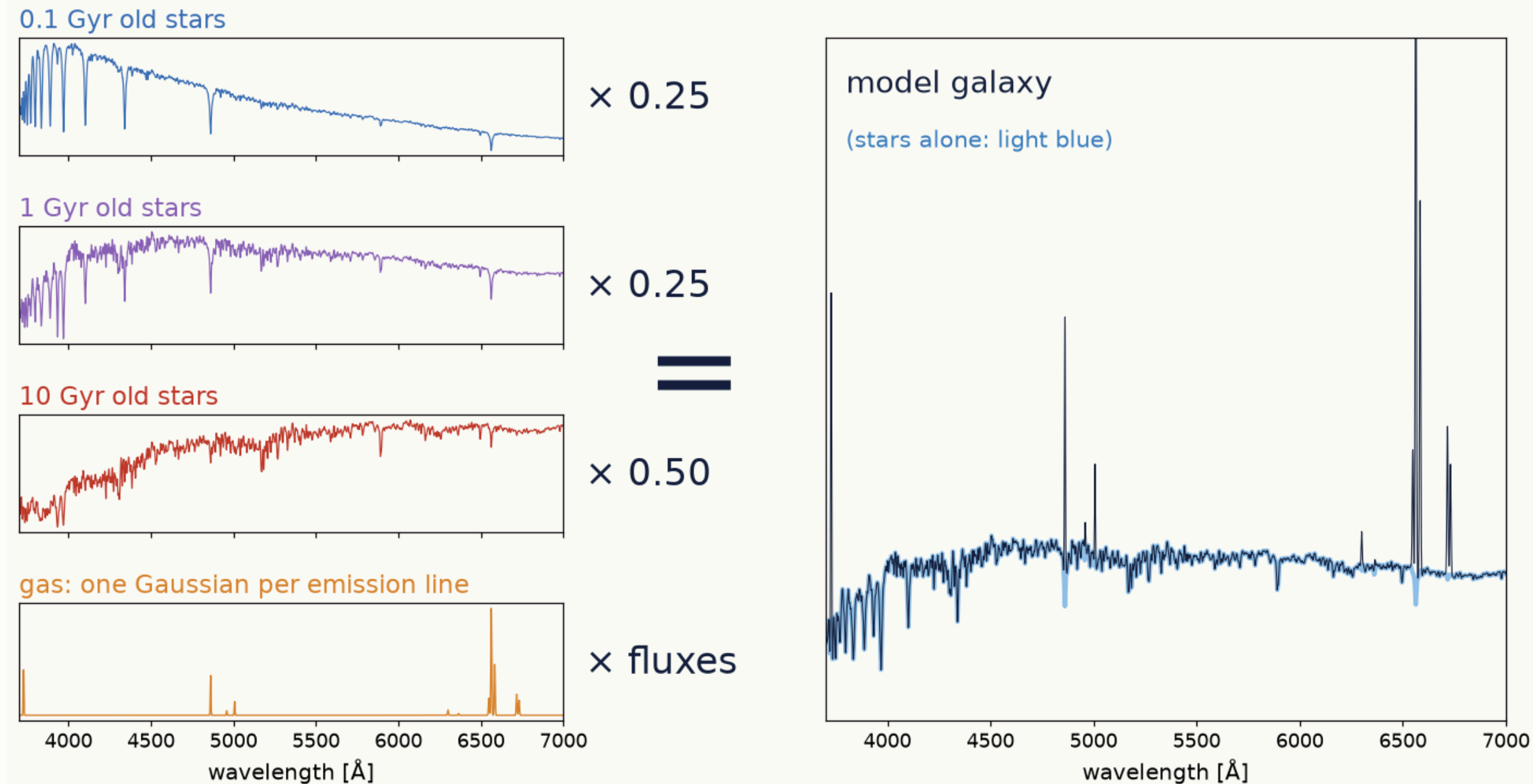
An SSP: stars all born at once, with the same metal content; many small stars, few massive ones.

Its spectrum: spectra of real stars, added up in the right numbers for that age.

As it ages: the blue fades, Balmer lines peak at 0.3–1 Gyr, the 4000 Å break and metal lines grow.

A galaxy made stars over time: its light is a **sum of SSPs**.

The pPXF model: templates $\times$ weights



Stellar templates: spectra of populations of one age and metallicity. pPXF uses 150.

Weights ≥ 0 : the fraction of the light each one gives.

Every template is then **shifted and broadened** by the motions (V, σ) and **reddened** by dust.

Gas templates: one Gaussian per line; their weights are the line fluxes.

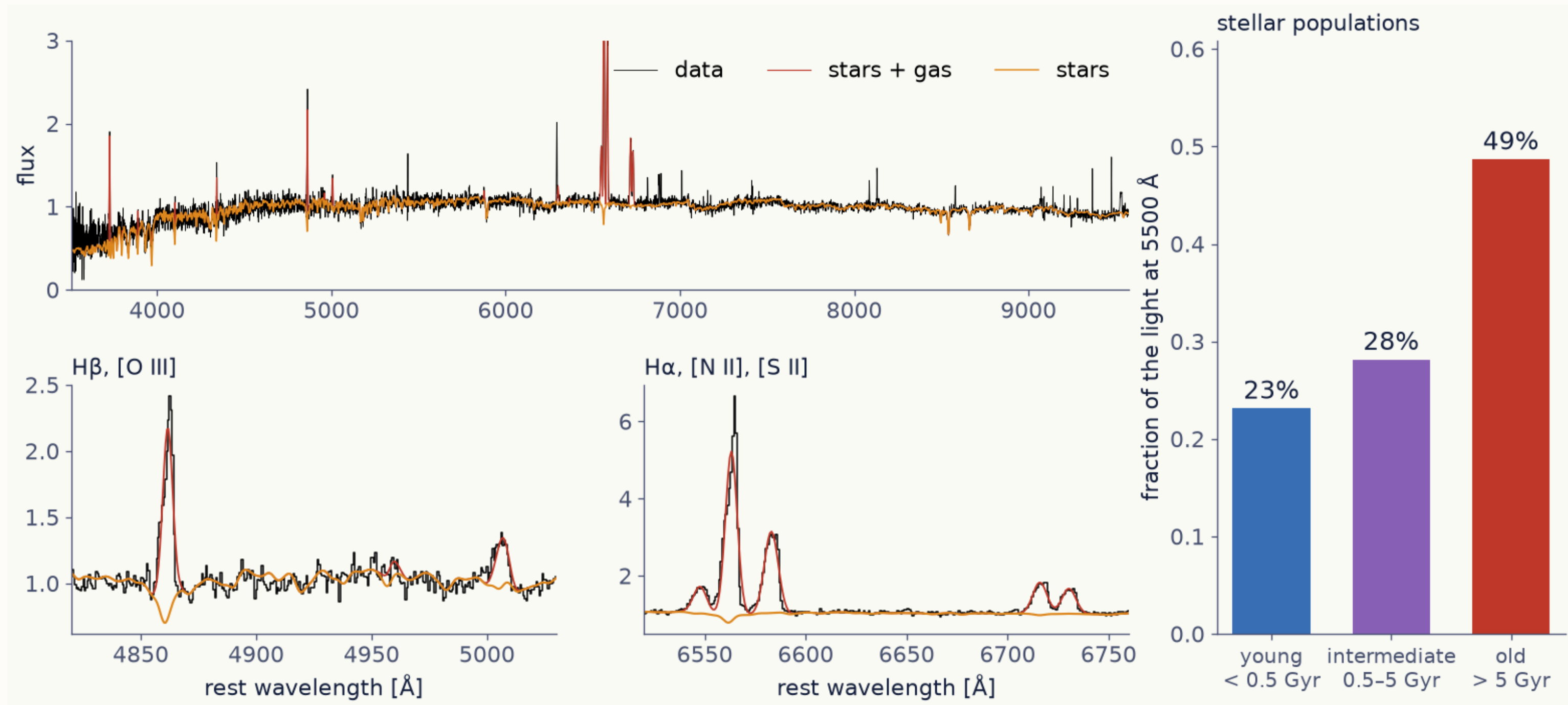
The pPXF fit



↻ Adjust V, σ and A_V and repeat from step 2 until χ^2 stops improving (pPXF takes care of this).

Output: the stellar populations (weights), **the line fluxes (gas weights)**, the kinematics and the dust.

pPXF on the nucleus of IC 3147B



$\chi^2/N \approx 2 \cdot \sigma_{\star} \approx 80$ km/s, gas ≈ 130 – 150 km/s · a mix of young and old stars · the four BPT lines come out as gas weights.

Now: open the notebook

TODAY · L1_SPECTRA_AND_PPXF

1.1–1.2 · pPXF's ingredients: star and gas templates;
set the emission lines yourself

1.3 · watch pPXF fit a real nucleus

Appendices A–C · spectral features, motions, dust: to
read at home

TOMORROW · L2_FROM_ARCHIVE_TO_BPT

Warm-up: build galaxies by hand

Choose a system and load its spectra

Fit them with pPXF and measure the lines

Better barred or interacting? Place them on the BPT

Choose your galaxy before lecture 2: [the sample gallery](#). (gallery.html in the course repository) shows every system, its companion and the spectrum of its nucleus.

Run in the browser: github.com/EdoBorsato/INTERBARS_classes → Binder or Colab